

JeevSetu

Multilingual, Offline-First Livestock Health Surveillance & Outbreak Early-Warning Platform

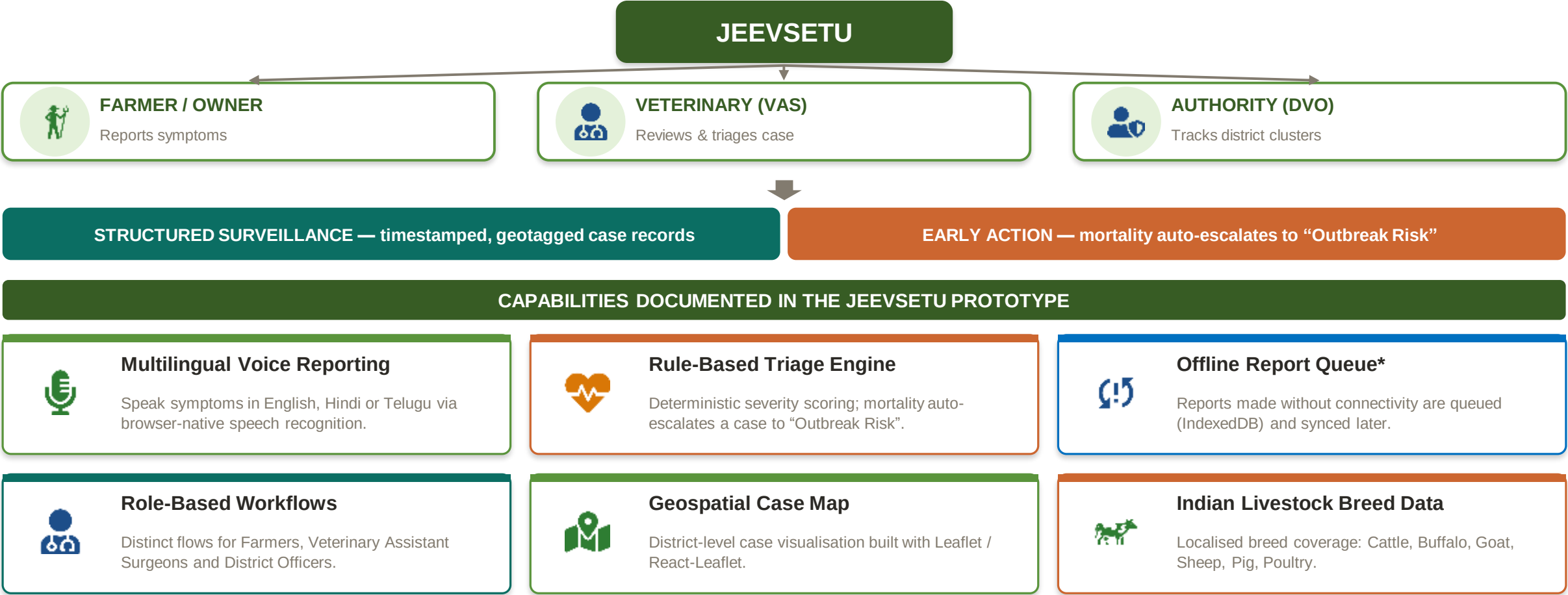
- **Problem Statement ID** – SIH26128
- **Problem Statement Title**- Efficient systems for early detection, prevention, and management of livestock diseases and animal health issues
- **Theme**- MedTech / BioTech / HealthTech
- **PS Category**- Software
- **Team ID**- AUS_SIH26_130
- **Team Name (Registered on portal)** – ChaoTics

Government of Maharashtra — Maharashtra State Innovation Society,
Dept. of Skills, Employment, Entrepreneurship & Innovation



IDEA TITLE

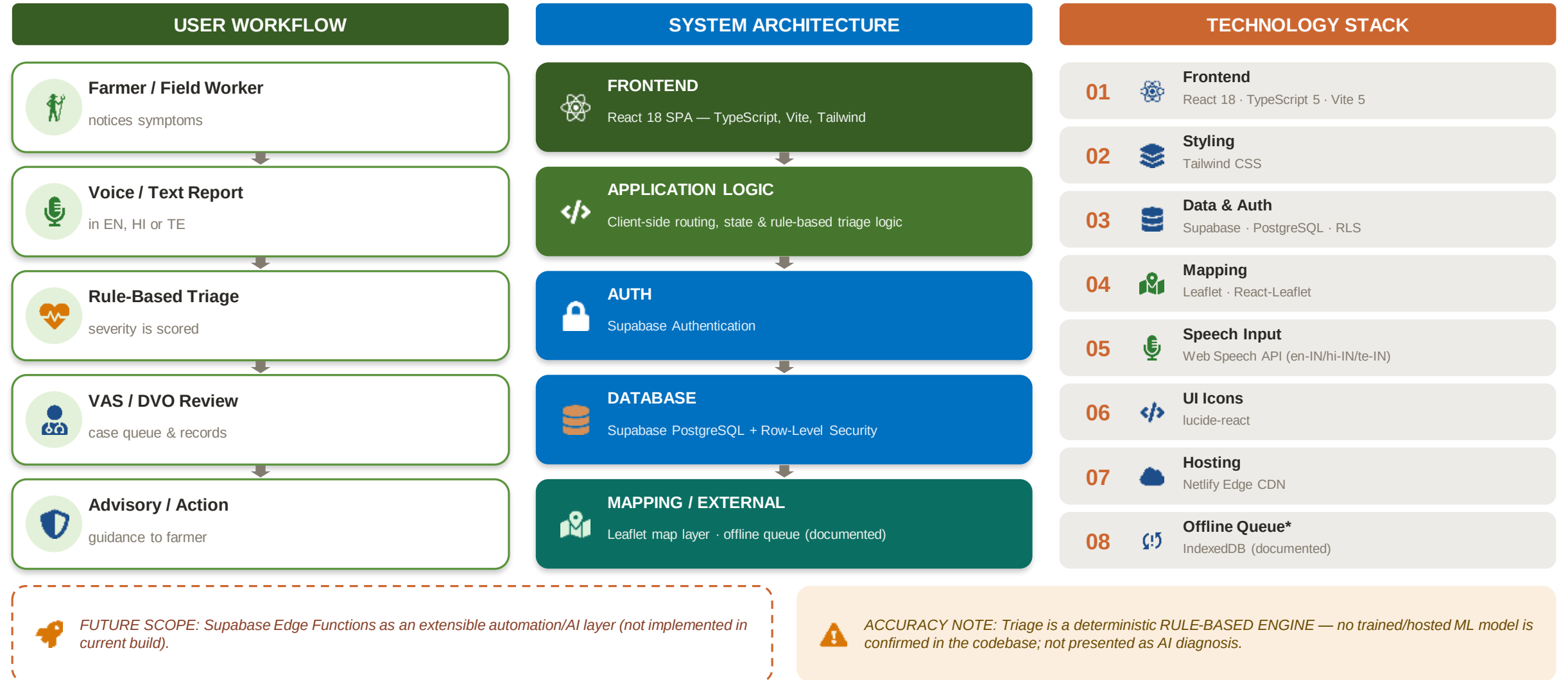
PROBLEM: Sick/dying livestock go unreported for days — language barriers, low literacy, patchy connectivity ► **OUTCOME:** faster escalation & district-level disease visibility



*Documented in project README; not independently verified from application source code.



TECHNICAL APPROACH





FEASIBILITY AND VIABILITY



CHALLENGE

**Rural connectivity is patchy or absent**

**Language & literacy barriers among field-level farmers**

**Delayed, informal disease reporting**

**Fragmented, paper-based surveillance across villages**

**Overloaded veterinary workforce**

JEEVSETU RESPONSE

Offline-First Reporting
Reports queue safely in IndexedDB and auto-sync once back online.

Multilingual Voice Input
Native dictation in English, Hindi or Telugu via the browser — zero cost.

Structured Reporting + Rule-Based Triage
Every report gets an instant, deterministic severity score.

Centralised Records + Outbreak Map
PostgreSQL records plus a geospatial heatmap with automatic 3-case village triggers.

Prioritised Case Queue
VAS case queue (Open → In Progress → Closed) with one-click lab sample requests.

WHY IT'S FEASIBLE

 No dependency on continuous connectivity for core reporting.

 No paid STT service and no app-store literacy hurdle.

 Auditable rules a vet can verify — not a opaque model.

 One source of truth per district; scales on managed Postgres.

 Existing VAS / DVO roles map directly onto the app.



TECHNICAL
React 18 + TypeScript 5 builds and type-checks today (npm run build / typecheck).



DEPLOYMENT
Already live on Netlify Edge CDN; Supabase manages Auth & Postgres — no server ops.



SCALABILITY
Stateless SPA + managed Postgres with RLS scales without re-architecture.

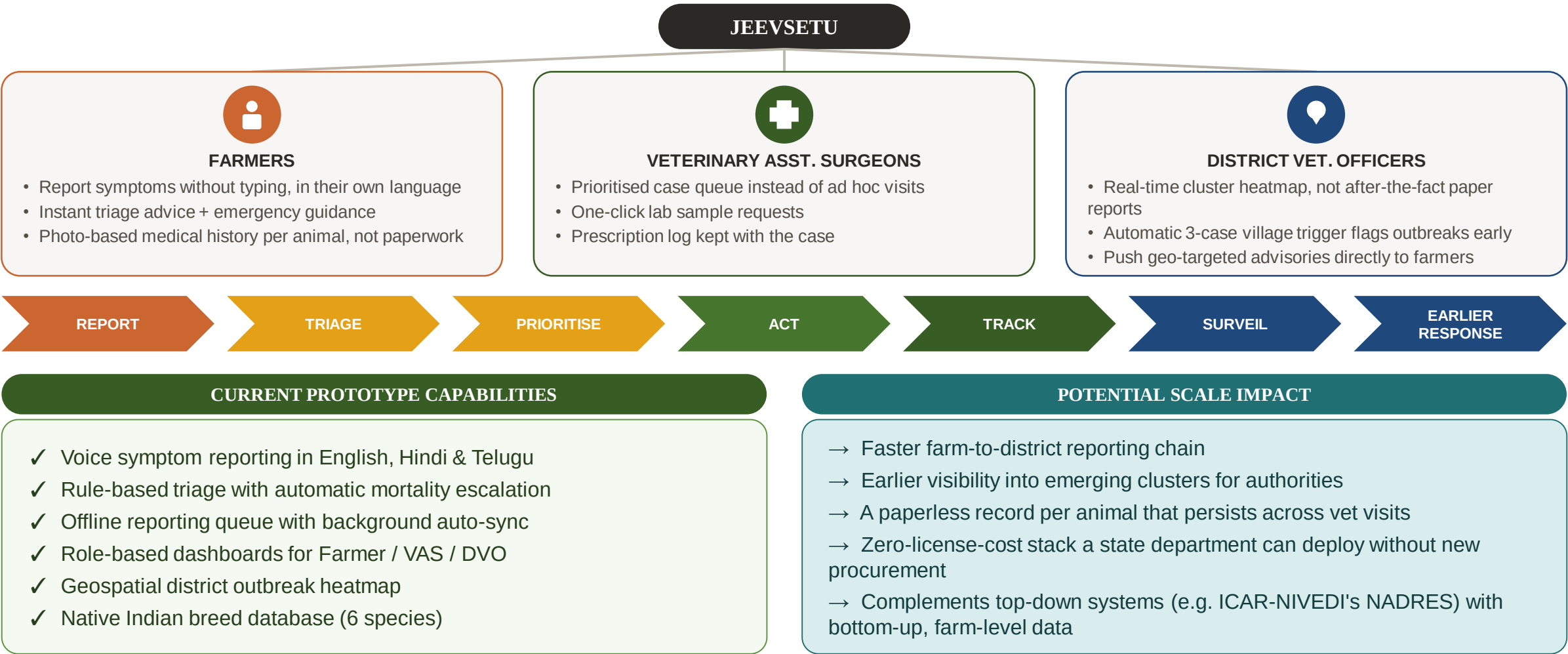


USABILITY
Voice-first in 3 Indian languages + offline queue removes literacy/connectivity hurdles.



OPERATIONAL
Farmer → VAS → DVO workflow mirrors the existing veterinary chain of command.

Why this matters: Foot-and-Mouth Disease alone costs India an estimated ₹20,000+ crore every year¹ — and losses compound the longer an outbreak goes unreported.





RESEARCH AND REFERENCES



01 GOVERNMENT / POLICY

National Animal Disease Control Programme (NADCP) — Dept. of Animal Husbandry & Dairying, GoI — dahd.gov.in/schemes/programmes/nadcp

20th Livestock Census 2019, All-India Report — Dept. of Animal Husbandry & Dairying, GoI — 535.78 million livestock nationwide

Problem Statement 26128 — Govt. of Maharashtra, Maharashtra State Innovation Society (MSInS)

02 DISEASE SURVEILLANCE

NADRES (v2/v3) — ICAR-NIVEDI — AI-driven district-level forewarning for 15 diseases — nivedi.res.in

Govindaraj et al. (2021) — Economic impact of FMD in cattle & buffaloes in India — Prev. Vet. Med. 190

Naidu et al. (2025) — Economic burden of Lumpy Skin Disease in India — Scientific Reports

03 SCIENTIFIC / INSTITUTIONAL

Singh et al. (2022) — FMD status in India, 2011–2020 — PMC / Indian J. Microbiol.

Rout et al. — Animal disease surveillance: importance & present status in India — Indian J. Med. Res.

04 TECHNOLOGY DOCUMENTATION

React 18 — react.dev

TypeScript 5 — typescriptlang.org

Vite 5 & Tailwind CSS — vitejs.dev · tailwindcss.com

Supabase (Auth, Postgres, RLS, Edge Functions) — supabase.com/docs

Web Speech API & IndexedDB — MDN Web Docs — developer.mozilla.org

05 MAPPING / WEB / INFRASTRUCTURE

Leaflet & React-Leaflet — leafletjs.com · react-leaflet.js.org

Netlify Edge CDN — netlify.com

Project repository — github.com/GavaraNeha/JeevSetu_SIH

Live prototype — deployed build on Netlify

(<https://6a95aa3455b4ec489d920721--remarkable-palmier-2eaa81.netlify.app/>)